

Unit 6.4 The cross product

Note Title

2012/05/23

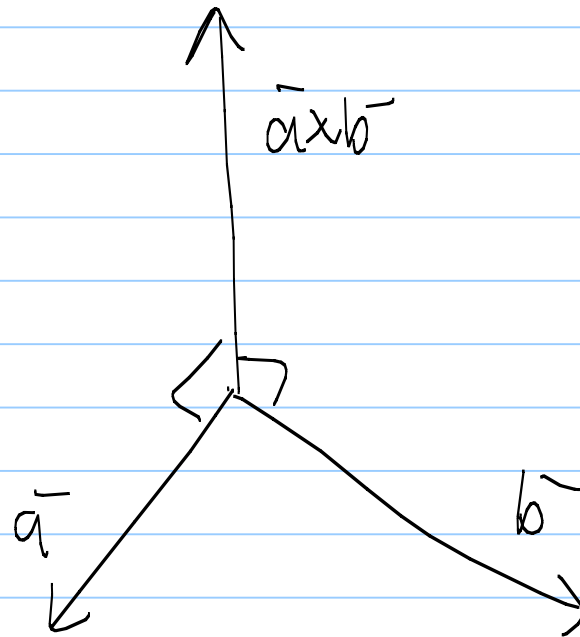
Stewart 12.4 p 808-810 and 812.

Definition If $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and

$\vec{b} = \langle b_1, b_2, b_3 \rangle$ then the cross product of $\vec{a}$ and $\vec{b}$ is the vector.

$$\vec{a} \times \vec{b} = \langle a_2 b_3 - a_3 b_2, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1 \rangle$$

Note: $\vec{a} \times \vec{b}$ is a vector.



- $\vec{a} \times \vec{b}$ is only defined when $\vec{a}$ and $\vec{b}$ is in $\mathbb{R}^3$ (three dimensions)

- Determinant of order 2

- $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

• Determinant of order 3

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$$

Then The cross product of two vectors
in $\mathbb{R}^3$

$$\vec{a} \times \vec{b} = \langle a_1, a_2, a_3 \rangle \times \langle b_1, b_2, b_3 \rangle$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$= \vec{i} \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \vec{j} \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \vec{k} \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}$$

Examples Find $\vec{a} \times \vec{b}$ if

$$\vec{a} = \langle 1, 2, 0 \rangle \text{ and } \vec{b} = \langle 3, -1, 2 \rangle$$

Sol:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 0 \\ 3 & -1 & 2 \end{vmatrix}$$

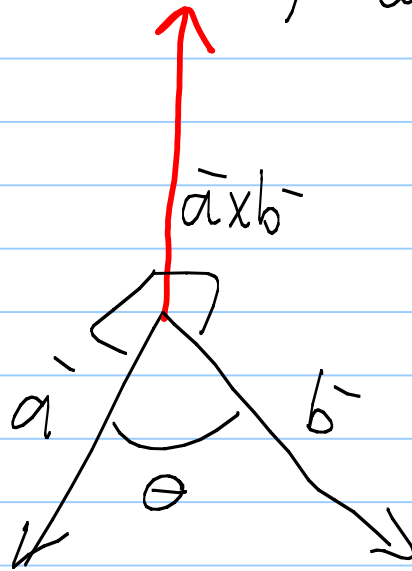
$$= \hat{i} \begin{vmatrix} 2 & 0 \\ 1 & 2 \end{vmatrix} - \hat{j} \begin{vmatrix} 1 & 0 \\ 3 & 2 \end{vmatrix} + \hat{k} \begin{vmatrix} 1 & 2 \\ 3 & -1 \end{vmatrix}$$

$$= \hat{i} (4 - 0) - \hat{j} (2 - 0) + \hat{k} (-1 - 6)$$

$$= 4\hat{i} - 2\hat{j} - 7\hat{k}$$

Theorem. Let $\vec{a}$ and $\vec{b}$ be vectors in $\mathbb{R}^3$. Let $\vec{a} \neq \vec{0}$ and $\vec{b} \neq \vec{0}$.

- The vector $\vec{a} \times \vec{b}$ is orthogonal (perpendicular) to both $\vec{a}$ and $\vec{b}$.



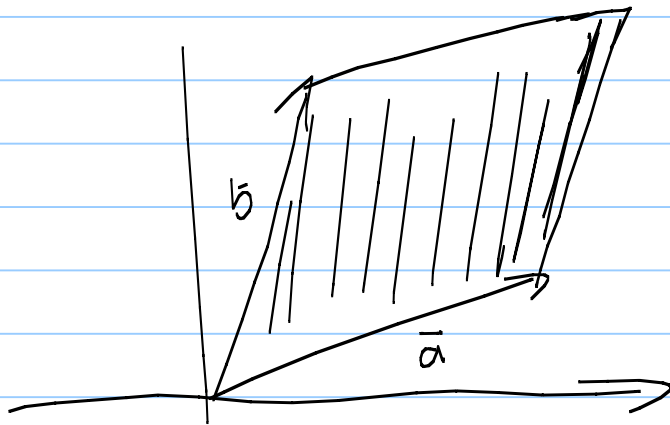
Corollary

Two nonzero vectors $\vec{a}$ and $\vec{b}$ are parallel if and only if

$$\vec{a} \times \vec{b} = \vec{0}$$

Note:

$|\vec{a} \times \vec{b}|$ is the area of the parallelogram determined by non-zero vectors $\vec{a}$ and $\vec{b}$.



Example Let $\vec{a} = \langle 2, 3, -1 \rangle$ and
 $\vec{b} = \langle 4, 6, -2 \rangle = 2\vec{a}$

Find $\vec{a} \times \vec{b}$. What can you conclude?

Sol:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -1 \\ 4 & 6 & -2 \end{vmatrix}$$

$$= \hat{i} \begin{vmatrix} 3 & -1 \\ 6 & -2 \end{vmatrix} - \hat{j} \begin{vmatrix} 2 & -1 \\ 4 & -2 \end{vmatrix} + \hat{k} \begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix}$$

$$= \hat{i} (-6 - (-6)) - \hat{j} (-4 - (-4)) + \hat{k} (12 - 12)$$

$$= \hat{i} (0) - \hat{j} (0) + \hat{k} (0) = \vec{0} \quad \therefore \vec{a} \text{ and } \vec{b} \text{ are parallel.}$$

Example. Find a vector of length 2 that is perpendicular to both

$$\vec{a} = \langle 1, -1, 2 \rangle \quad \text{and} \quad \vec{b} = \langle 3, 2, -1 \rangle.$$

Sol: $\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -1 & 2 \\ 3 & 2 & -1 \end{vmatrix}$

$$= \vec{i} \begin{vmatrix} -1 & 2 \\ 2 & -1 \end{vmatrix} - \vec{j} \begin{vmatrix} 1 & 2 \\ 3 & -1 \end{vmatrix} + \vec{k} \begin{vmatrix} 1 & -1 \\ 3 & 2 \end{vmatrix}$$

$$= \bar{i}(1-4) - \bar{j}(-1-6) + \bar{k}(2-(-3))$$

$$= -3\bar{i} + 7\bar{j} + 5\bar{k}$$

Find unit vector

$$\bar{u} = \langle -3, 7, 5 \rangle \cdot \frac{1}{\sqrt{(-3)^2 + (7)^2 + (5)^2}}$$

Vector of length 2

$$2\bar{u} = \left\langle \frac{-6}{\sqrt{83}}, \frac{14}{\sqrt{83}}, \frac{10}{\sqrt{83}} \right\rangle$$

Example. Find The area of the triangle with vertices

$A(2, -3, 4)$, $B(1, 2, 3)$ and $C(0, -2, -1)$

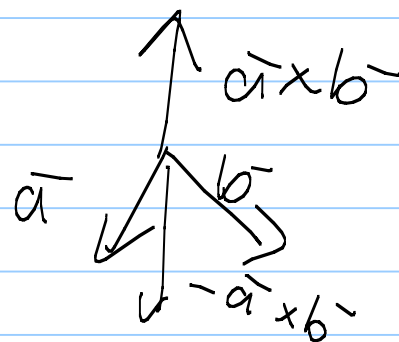
Sol:

Find $\overrightarrow{AB}$ and $\overrightarrow{AC}$

$$\text{area of The triangle} = \frac{1}{2} \left| \overrightarrow{AB} \times \overrightarrow{AC} \right|$$

Properties: Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be vectors in $\mathbb{R}^3$ and $c \in \mathbb{R}$.

1. $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$



2. $(c\vec{a}) \times \vec{b} = c(\vec{a} \times \vec{b})$

and

$$\vec{a} \times (c\vec{b}) = c(\vec{a} \times \vec{b})$$

$$3. \bar{a} \times (\bar{b} + \bar{c}) = \bar{a} \times \bar{b} + \bar{a} \times \bar{c}$$

$$4. (\bar{a} + \bar{b}) \times \bar{c} = \bar{a} \times \bar{c} + \bar{b} \times \bar{c}$$

Example

Find a vector perpendicular to the plane that passes through the points $P(1, -1, 1)$, $Q(2, 0, 1)$ and $R(2, -1, 1)$

